Sexy-Prime Bridges Across Goldbach Transitions

Exact Convolutions, Modular Extinction, Local Sieve Dynamics, and Conditional Covering Criteria

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Abstract

We investigate a proposed rescue mechanism for Goldbach representations at indices $n \equiv 1 \pmod 3$ where the exceptional Goldbach summand 3 is unavailable. The proposed rescue uses a prime pair $(q, q+6)$ and a complementary prime r, so that $q + r = 2n - 4$ and $(q + 6) + r = 2n + 2$. Writing $n = 3t + 1$ reduces the problem to positivity of the discrete convolution $R(t) = \sum_{a=1}^{t-1} f(a)f(a+1)f(t-a)$, where $f(a) = \mathbf{1}_{\mathbb{P}}(6a - 1)$. We give exact gcd identities, the associated singular series and its uniform positivity, elementary rough-number bounds, finite local-covering certificates, and infinite families of prescribed extinction and recovery. A conditional covariance criterion isolates the additional additive information required to prove universal rescue. Exact modular convolutions verify $R(t) > 0$ for $2 \le t \le 10^6$. The universal rescue assertion and the infinitude of sexy-prime pairs remain open; the note does not claim a solution of either problem.

Status and conventions. Statements designated *theorem* or *proposition* below have proofs; computational statements have explicitly finite scopes; conjectures and conditional results are labeled as such. The word "sexy" denotes a prime gap of exactly six, not necessarily consecutive primes.

1 Goldbach intersections and the extinction condition

Let $\mathbb{P}$ be the set of positive rational primes. For $n \ge 2$ define

$$S(n) = \{p \in \mathbb{P} : 2n - p \in \mathbb{P}\}, \qquad I_k(n) = S(n) \cap S(n-k), \quad n > k. \tag{1}$$

Both members of each Goldbach pair belong to $S(n)$. The identity

$$I_3(n) \neq \varnothing \iff \exists\, p, q \in \mathbb{P} : \quad p + q = 2n - 6, \quad q + 6 \in \mathbb{P} \tag{2}$$

is immediate, since $p + q = 2n - 6$ and $p + (q + 6) = 2n$. The gap-six intersection property has been discussed and experimentally tested previously; see [8].

Proposition 1.1 (Direct modular extinction)**.** *Let $n \ge 7$, $n \equiv 1 \pmod 3$, and suppose that $2n-3$ is composite. For every Goldbach representation $2n = p+q$, both p and q are $1 \pmod 3$. Consequently, $p + 2$ and $q + 2$ are composite.*

Proof. The residue of $2n$ modulo 3 is 2. If a summand equals 3, the other is $2n - 3$, contrary to the assumption that it is prime. Thus both primes are nonzero modulo 3, and the only pair of nonzero residues adding to 2 is $(1, 1)$. Each increment by 2 is divisible by 3 and greater than 3. □

The failure of this particular continuation does *not* mean Goldbach fails at $2n + 2$. We study a different, six-unit bridge that may rescue the transition.

Definition 1.2 (Sexy-prime rescue)**.** At an index $n \equiv 1 \pmod 3$, a *backward sexy-prime rescue* is a triple of primes $q, q+6, r$ such that

$$2n-4 = q+r, \qquad 2n+2 = (q+6)+r. \tag{3}$$

The two Goldbach targets are separated by six. Such a triple yields $I_3(n+1) \neq \varnothing$.

For $n \equiv 1 \pmod 3$, if $r \neq 3$ in (3), then $q \equiv r \equiv 5 \pmod 6$. There can also be exceptional representations with $r = 3$ and $q \equiv 1 \pmod 6$; our main count deliberately restricts to the nonexceptional orientation.

2 Exact convolution and the central conjecture

Write

$$n = 3t+1, \quad M = 2n-3 = 6t-1, \quad f(a) = \mathbf{1}_{\mathbb{P}}(6a-1), \quad g(a) = f(a)f(a+1) \tag{4}$$

for positive integers a. A lower sexy prime in the relevant orientation is $q = 6a-1$, with successor $q+6 = 6a+5$. Its complementary candidate in (3) is

$$r = M - 6a = 6(t-a) - 1. \tag{5}$$

Therefore the exact count is

$$\boxed{R(t) = \sum_{a=1}^{t-1} f(a)f(a+1)f(t-a) = (g * f)(t).} \tag{6}$$

Here the convolution is over positive indices; the upper limit makes all three prime arguments positive.

Conjecture 2.1 (Extinction rescue)**.** There is t_0 such that, for every $t \geq t_0$,

$$f(t) = 0 \quad \Longrightarrow \quad R(t) > 0. \tag{7}$$

A stronger form asks for $R(t) > 0$ for all sufficiently large t, without restricting to extinction indices.

Remark 2.2. Conjecture 2.1 would imply the infinitude of sexy-prime pairs. Indeed, if only finitely many eligible a had $g(a) = 1$, then $R(t) > 0$ on a set of positive density would force at least one of finitely many translates of the ordinary primes to cover a positive-density set of indices, contrary to the prime number theorem. Thus even the weaker rescue conjecture lies beyond what is currently known.

3 Divisibility identities peculiar to extinction

Proposition 3.1 (Factor transfer)**.** *For $M = 6t-1$ and $q = 6a-1$,*

$$\boxed{\gcd(M-6a, M) = \gcd(a, M).} \tag{8}$$

Thus for every prime $\ell \mid M$,

$$\ell \mid (2n-4-q) \quad \Longleftrightarrow \quad \ell \mid a = \frac{q+1}{6}. \tag{9}$$

Proof. $\gcd(M-6a, M) = \gcd(6a, M) = \gcd(a, M)$ because $M \equiv -1 \pmod 6$. □

Corollary 3.2 (The exceptional modulus five)**.** *If $g(a) = 1$, then $5 \nmid a$. Hence if $5 \mid M$, no eligible sexy-prime complement $M - 6a$ is divisible by 5.*

Proof. If $5 \mid a$, then $6a+5$ is a multiple of 5 exceeding 5, so $f(a+1) = 0$. Apply Proposition 3.1. □

A complementary exact description is obtained from the three linear forms

$$L_1(a) = 6a - 1, \qquad L_2(a) = 6a + 5, \qquad L_3(a) = 6(t - a) - 1. \tag{10}$$

For $\ell \geq 5$ with $\ell \mid M$, the forbidden residues for a are $t, t-1, 0 \pmod \ell$ respectively (since $6t \equiv 1 \pmod \ell$). For $\ell = 5$ two collide; for $\ell \geq 7$ they are distinct. Consequently the fraction of residue classes compatible with the first two forms that also survive the third is

$$\begin{cases} 1, & \ell = 5, \\ (\ell - 3)/(\ell - 2), & \ell \geq 7. \end{cases} \tag{11}$$

Multiplying the fractions gives the *local model*

$$D(M) = \prod_{\substack{\ell \mid M \\ \ell \geq 7}} \frac{\ell - 3}{\ell - 2} \gg \frac{1}{\log\log(3M)}. \tag{12}$$

For the last inequality, compare each factor to $(1 - 1/\ell)(1 - O(\ell^{-2}))$ and use the standard lower bound for $\varphi(M)/M$. Equation (12) concerns locally admissible residues; it is *not* a lower bound on the number of actual sexy primes surviving the factors of M.

For completeness, the elementary Möbius identity

$$\#\{1 \leq a \leq A : \gcd(a, M) = 1\} = A\frac{\varphi(M)}{M} + O(2^{\omega(M)}) \tag{13}$$

shows that $\gg t/\log\log t$ index values $a < t$ are coprime to M. They need not satisfy $g(a) = 1$.

4 The singular series: positivity and its limits

Let $\nu_\ell(t)$ be the number of residue classes $a \pmod \ell$ annihilating at least one of (10). For $\ell = 2, 3$, the three forms are nonzero in every residue class, and $\nu_\ell(t) = 0$. For $\ell \geq 5$, their zeros are

$$a \equiv 6^{-1}, \quad -5 \cdot 6^{-1}, \quad t - 6^{-1} \pmod \ell. \tag{14}$$

The first two never coincide; the last coincides with one of them precisely when $\ell \mid (6t-2)(6t+4)$. Hence

$$\nu_\ell(t) = \begin{cases} 0, & \ell \in \{2, 3\}, \\ 2, & \ell \geq 5,\ \ell \mid (6t-2)(6t+4), \\ 3, & \ell \geq 5,\ \ell \nmid (6t-2)(6t+4). \end{cases} \tag{15}$$

Theorem 4.1 (Explicit singular series)**.** *The singular series of the three-form system is*

$$\boxed{\mathfrak{S}(t) = 27C_3 \prod_{\substack{\ell \geq 5 \\ \ell \mid (6t-2)(6t+4)}} \frac{\ell - 2}{\ell - 3}, \qquad C_3 = \prod_{\ell \geq 5} \frac{1 - 3/\ell}{(1 - 1/\ell)^3} > 0.} \tag{16}$$

In particular, $\mathfrak{S}(t) \geq 27C_3$ uniformly in t.

Proof. By definition, $\mathfrak{S}(t) = \prod_\ell (1 - \nu_\ell(t)/\ell)/(1 - 1/\ell)^3$. The factors for 2 and 3 multiply to $8 \cdot (27/8) = 27$. For $\ell \geq 5$ with three distinct zeros the factor is that in C_3; when two zeros coincide its ratio to the generic factor is $(\ell - 2)/(\ell - 3)$. Since the generic factors are $1 + O(\ell^{-2})$, C_3 converges to a strictly positive number. □

The Hardy–Littlewood heuristic [7] suggests

$$R(t) \sim \mathfrak{S}(t) \frac{t}{(\log(6t))^3}, \tag{17}$$

uniformly enough in t to ensure positivity. This is *not* a proved asymptotic. Notably the enhancement factors in (16) depend on the two even Goldbach targets $6t - 2$ and $6t + 4$, not on the factorization of the extinction number $6t - 1$ itself.

5 An elementary lower bound for rough triples

An Erdős-style primorial estimate proves plentiful local survivors, although not primes. Let $N = 6t + 4$ and put

$$y = \left\lfloor \frac{\log N}{8 \log 4} \right\rfloor, \qquad P(y) = \prod_{\ell \leq y} \ell. \tag{18}$$

The elementary binomial-coefficient bound $P(y) \leq 4^{2y}$ implies $P(y) \leq N^{1/4}$; compare Erdős's proof of Bertrand's postulate [3]. Consider x in $[N/4, N/2]$ for which

$$\gcd(x(x + 6)(N - 6 - x), P(y)) = 1. \tag{19}$$

Proposition 5.1 (Simultaneous logarithmic roughness)**.** *For all sufficiently large even N, the number of such integers x is $\gg N/(\log N)^3$, with an absolute implicit constant.*

Proof. For $\ell = 2$ the three forms have one forbidden residue and for $\ell = 3$ at most two. For $\ell \geq 5$ there are at most three, hence the proportion of admissible residues modulo $P(y)$ is at least

$$\frac{1}{6} \prod_{5 \leq \ell \leq y} \left(1 - \frac{3}{\ell}\right) \geq \frac{1}{6} \prod_{k=5}^{\lfloor y \rfloor} \frac{k - 3}{k} = \frac{4}{m(m - 1)(m - 2)}, \quad m = \lfloor y \rfloor.$$

Counting complete periods inside an interval of length $N/4$ gives at least $ND/4 - P(y)$ survivors, where D is the displayed local density. Since $m \asymp \log N$ and $P(y) \leq N^{1/4}$, the result follows. □

With Mertens' theorem the lower bound improves to $\gg N/(\log \log N)^3$. These rough integers mostly include composites; sieving to $\sqrt{N}$ lies beyond the complete-period argument and encounters the usual sieve parity obstruction. The triple-divisor correlations of Goldston and Yıldırım [5] are relevant background, not a solution of this problem.

6 Exact local extinction and finite recovery

Fix distinct eligible lower sexy primes $Q = \{q_1, \ldots, q_m\}$, and let

$$P_z = \prod_{5 \leq \ell \leq z} \ell, \qquad S_z(t; Q) = \{q \in Q : \gcd(6t - 2 - q, P_z) = 1\}. \tag{20}$$

With $\ell > z$ the next prime,

$$S_\ell(t; Q) = S_z(t; Q) \setminus \{q \in S_z(t; Q) : q \equiv 6t - 2 \pmod{\ell}\}. \tag{21}$$

This is an exact, one-sided sieve transition, rather than a Markov approximation.

Let $H_z(Q)$ be the number of $t \pmod{P_z}$ with $S_z(t;Q) = \varnothing$. Let $C_{z,\ell}(Q)$ count those classes with nonempty $S_z(t;Q)$ whose elements all have the same residue modulo ℓ. The CRT gives the exact recurrence

$$\boxed{H_\ell(Q) = \ell H_z(Q) + C_{z,\ell}(Q).} \tag{22}$$

Old empty classes have ℓ empty extensions; a nonempty class can become empty at one extension exactly when its survivors lie in one class modulo ℓ.

Proposition 6.1 (Finite local-covering certificate)**.** *For*

$$Q_{16} = \{5, 11, 17, 23, 41, 47, 53, 83, 101, 107, 131, 167, 173, 191, 227, 233\},$$

we have

$$\min_{t \bmod P_{19}} |S_{19}(t; Q_{16})| = 1, \qquad H_{23}(Q_{16}) = 2,$$

where $P_{19} = 1\,616\,615$ *and* $P_{23} = 37\,182\,145$*. The two complete-extinction classes modulo* P_{23} *are*

$$t \equiv 24\,004\,568, \quad 34\,450\,388 \pmod{P_{23}}.$$

Proof. This is an exhaustive finite CRT calculation. A reproducible bit-mask program is supplied as `verify_local.py` in the source package. The minimum-one classes modulo P_{19} are $t \equiv 501473$ (only $q = 107$ survives) and $t \equiv 1371958$ (only $q = 233$ survives); the single additional root class modulo 23 in each case gives the two listed extinctions. □

Remark 6.2. A fixed finite Q can always be made completely extinct by choosing distinct large primes ℓ_i and imposing $6t - 2 - q_i \equiv 0 \pmod{\ell_i}$, using CRT. This does not imply that the *full* family of sexy primes fails at that index.

7 Infinite recovery after prescribed finite failures

Theorem 7.1 (Prescribed rescuer after finite extinction)**.** *Let* $q_1, \ldots, q_m, q_*$ *be distinct eligible lower sexy primes. There are infinitely many extinction indices* $n \equiv 1 \pmod 3$ *at which every* q_i *fails but* q_* *succeeds.*

Proof. Choose distinct primes $\ell_0, \ell_1, \ldots, \ell_m > 3$ so large that $\ell_0 > q_* + 1$ and $\ell_i > |q_i - q_*|$ for $i \geq 1$. By CRT choose n in one arithmetic progression satisfying

$$n \equiv 1 \pmod 3, \quad 2n - 3 \equiv 0 \pmod{\ell_0}, \quad 2n - 4 - q_i \equiv 0 \pmod{\ell_i}.$$

For sufficiently large n, this enforces genuine compositeness of the extinction number and of all old complements. Along that progression the intended new complement $r_*(n) = 2n - 4 - q_*$ is coprime to its common difference: modulo ℓ_0 it is $-1 - q_*$, modulo ℓ_i it is $q_i - q_*$, and it is coprime to 2 and 3. Dirichlet's theorem supplies infinitely many prime values of $r_*(n)$. □

A stronger simultaneous-recovery statement follows from the published DHL$^*(50;2)$ theorem for admissible families of affine forms [2, 4]. It is important here not to assume the existence of infinitely many sexy primes.

Corollary 7.2 (Two later rescuers after the first sixteen fail)**.** *There are infinitely many extinction indices at which all members of* Q_{16} *fail and at least two distinct eligible sexy primes outside* Q_{16} *succeed.*

Proof. Choose 50 explicitly existing sexy primes $q > 233$ (for example, the first 50 in this class; their existence is a finite primality check). Impose composite obstructions on Q_{16} and on $2n-3$ using distinct auxiliary primes all larger than the 50 chosen primes and than 233. Along the resulting CRT progression, the 50 complementary affine forms are nonzero modulo each auxiliary prime, and modulo 2 and 3. At any prime $5 \le \ell \le 50$ not dividing the progression's common difference, the residue class corresponding to $q \equiv 0 \pmod{\ell}$ is absent, since all 50 chosen q are prime and greater than 50. For $\ell > 50$ at most 50 classes are forbidden. Thus the affine forms are admissible. The DHL$^*(50;2)$ theorem then provides two primes simultaneously for infinitely many parameters, yielding the claim. □

This infinite-family result proves *possibility of recovery*, not recovery at every index. Indeed, no finite set of rescuer pairs covers all sufficiently large extinction indices.

8 Union estimates and additive complements

For a fixed eligible q, put

$$A_q = \{n \equiv 1 \ (3) : \ 2n-3 \text{ composite}, \quad 2n-4-q \in \mathbb{P}\}.$$

Proposition 8.1 (Finite unions of rescue sets)**.** *For any fixed distinct eligible $q_1, \dots, q_m$,*

$$\#\left([1,X] \cap \bigcup_{i=1}^{m} A_{q_i}\right) \sim \frac{mX}{\log X}. \tag{23}$$

Proof. For each q, the complementary prime lies in the class 5 (mod 6) and ranges up to $2X + O_q(1)$; the prime number theorem in arithmetic progressions gives $X/\log X$ choices. The indices for which $2n-3$ is also prime are $O_q(X/\log^2 X)$ by the usual fixed-difference upper-bound sieve. The overlap of two distinct rescue sets imposes a pair of primes with fixed nonzero difference $q_i - q_j$, again numbering $O_{q_i,q_j}(X/\log^2 X)$. Bonferroni's inequalities give (23) for fixed m. □

The word *fixed* is essential: the error constants are not uniform enough to let m grow into the covering range. The recent conditional Romanov–Selberg analysis of twin-prime Goldbach representations in [9] is a close analogue of this distinction between positive-density coverage and full coverage.

Proposition 8.2 (A necessary abundance condition)**.** *If Conjecture 2.1 holds, then the number $m(Y) = \#\{q \le Y : q \equiv 5 \ (6), \ q, q+6 \in \mathbb{P}\}$ obeys*

$$m(Y) \gg \log Y. \tag{24}$$

Moreover, applying the additive-complement theorem announced in [10] would strengthen this to

$$m(Y) \gg \log Y \log\log Y. \tag{25}$$

Proof. For the elementary bound, the number of extinction indices $n \le X$ is $X/3 + O(X/\log X)$. Their rescue representations use one $q \le 2X$ and one complementary prime $r \le 2X$, whence $m(2X)\pi(2X) \gg X$. The prime number theorem gives $m(2X) \gg \log X$.

For the second assertion, let

$$\mathcal{B} = \{0, 1, \dots, 5\} \ \cup \ \{q + 1 + j : q \in \mathcal{Q}, \ 0 \le j \le 5\}.$$

For every sufficiently large integer K, choose $j \in \{0, \dots, 5\}$ such that $T = K - j \equiv 5 \pmod 6$. If T is prime, $K = T + j \in \mathbb{P} + \mathcal{B}$. Otherwise set $n = (T+3)/2$, which is an extinction

index, and choose a rescue $T = r + (q+1)$. Thus $K = r + (q+1+j) \in \mathbb{P} + \mathcal{B}$. So $\mathcal{B}$ is an asymptotic additive complement to $\mathbb{P}$. The theorem of [10] yields $B(Y) \gg \log Y \log\log Y$; since $B(Y) \le 6 + 6m(Y)$, (25) follows. □

The new October 2026 result [10] is a preprint. Neither (24) nor (25) reverses into universal coverage.

9 Higher-order intersections and lower-tail covering

The preceding union estimate cannot be extended to a growing family by assuming independence. For a finite set Q of eligible rescuers, write $X_z(t) = |S_z(t;Q)|$ and average t over the entire residue system modulo P_z. Put

$$V_z = \prod_{5 \le \ell \le z} \left(1 - \frac{1}{\ell}\right), \qquad m = |Q|.$$

For a nonzero difference d define the finite Euler product

$$C_z(d) = \prod_{5 \le \ell \le z} \begin{cases} \dfrac{\ell}{\ell - 1}, & \ell \mid d, \\ 1 - \dfrac{1}{(\ell-1)^2}, & \ell \nmid d. \end{cases}$$

Proposition 9.1 (Exact first and second local moments)**.** *With $Q = \{q_1, \dots, q_m\}$ as above,*

$$\mathbb{E}X_z = mV_z, \tag{26}$$

$$\mathbb{E}X_z^2 = mV_z + V_z^2 \sum_{i \ne j} C_z(q_i - q_j). \tag{27}$$

Consequently the proportion of residue classes with at least one survivor satisfies

$$\Pr(X_z > 0) \ge \frac{m^2}{m/V_z + \sum_{i \ne j} C_z(q_i - q_j)}. \tag{28}$$

Proof. For each fixed q, one class modulo each ℓ is forbidden, so $\Pr(q \in S_z) = V_z$. For distinct q_i, q_j, their forbidden classes coincide modulo ℓ exactly when $\ell \mid q_i - q_j$. Thus their joint survival density is $V_z^2 C_z(q_i - q_j)$. Expanding X_z^2 gives the second moment. Cauchy–Schwarz yields $(\mathbb{E}X_z)^2 \le \mathbb{E}X_z^2 \Pr(X_z > 0)$. □

For a larger fixed subfamily $J \subseteq Q$, the exact simultaneous survival fraction is

$$\Pr(J \subseteq S_z(t;Q)) = \prod_{5 \le \ell \le z} \left(1 - \frac{\nu_\ell(J)}{\ell}\right), \quad \nu_\ell(J) = \#\{q \bmod \ell : q \in J\}. \tag{29}$$

It can vanish when the chosen rescuers occupy every residue class modulo a prime. Formula (29) describes *finite-sieve* intersections, not the joint distribution of actual complementary primes.

There is also a direct, one-sided criterion for *complete* coverage by a growing family. For $X \ge 1$ define

$$E_X = \{n \in [X, 2X] \cap \mathbb{Z} : n \equiv 1 \pmod 3,\ 2n-3 \text{ composite}\}, \quad R_B(n) = \sum_{\substack{q \le B \\ q \in \mathcal{Q}}} \mathbf{1}_{\mathbb{P}}(2n - 4 - q),$$

where $\mathcal{Q} = \{q \equiv 5 \pmod 6 : q, q+6 \in \mathbb{P}\}$ and $\mathbf{1}_{\mathbb{P}}(r) = 0$ for $r \le 1$. Set

$$Z_B(X) = \sum_{n \in E_X} e^{-R_B(n)}. \tag{30}$$

Every uncovered index contributes exactly 1 to $Z_B(X)$, so

$$\#\{n \in E_X : R_B(n) = 0\} \le Z_B(X). \tag{31}$$

Theorem 9.2 (Conditional exponential lower-tail criterion)**.** *Suppose there are constants $c, C > 0$ and cutoffs $B = B(X)$ such that*

$$\frac{Z_B(X)}{|E_X|} \le C \exp\left(-c \frac{m_B}{\log X}\right), \qquad m_B = \#(\mathcal{Q} \cap [1, B]), \tag{32}$$

for every sufficiently large X. If also $m_B \ge (2/c)(\log X)^2$, then every extinction index in $[X, 2X]$ has a rescuer $q \le B(X)$ for all sufficiently large X.

Proof. Equations (31) and (32) give

$$\#\{n \in E_X : R_B(n) = 0\} \le CX \exp(-2 \log X) = C/X < 1.$$

The count on the left is a nonnegative integer, hence zero. □

The exponential moment itself has the exact expansion

$$Z_B(X) = \sum_{J \subseteq \mathcal{Q} \cap [1,B]} (e^{-1} - 1)^{|J|} \# \left(E_X \cap \bigcap_{q \in J} A_q \right). \tag{33}$$

This identity makes clear where uniform higher-order prime-tuple information would have to enter. No estimate of the strength (32) is presently known for this family. In particular, finite-order Selberg upper-bound estimates do not yield it.

10 One-sided additive mixing and a conditional theorem

A more stringent, numerically supported conjecture restricts sexy-prime indices to the interior interval

$$J_t = [\lceil t/4 \rceil, \lfloor t/2 \rfloor] \cap \mathbb{Z}, \quad R_{\mathrm{mid}}(t) = \sum_{a \in J_t} g(a) f(t-a). \tag{34}$$

This avoids relying on extremely small rescue indices. Write

$$K_t = \sum_{a \in J_t} g(a), \qquad v_t(a) = \Lambda(6(t-a) - 1), \qquad \overline{v}_t = \frac{1}{|J_t|} \sum_{a \in J_t} v_t(a),$$

and define the unnormalized covariance

$$\mathcal{C}_t = \sum_{a \in J_t} \left(g(a) - \frac{K_t}{|J_t|} \right) (v_t(a) - \overline{v}_t).$$

The exact identity is

$$W_t := \sum_{a \in J_t} g(a) v_t(a) = K_t \overline{v}_t + \mathcal{C}_t. \tag{35}$$

The prime number theorem in the progression 5 (mod 6) gives $\overline{v}_t = 3 + o(1)$.

Theorem 10.1 (Conditional one-sided mixing criterion)**.** *Suppose for all sufficiently large t that*

$$\frac{K_t}{\sqrt{t}\log t} \longrightarrow \infty, \qquad \mathcal{C}_t \geq -(3-\eta)K_t \tag{36}$$

for some fixed $\eta > 0$. Then $R_{\mathrm{mid}}(t) > 0$ for every sufficiently large t, and hence Conjecture 2.1 holds.

Proof. Equation (35) and the assumptions imply $W_t \geq (\eta + o(1))K_t$. A nonprime with nonzero von Mangoldt weight is a proper prime power. There are $O(\sqrt{t})$ such numbers among the values $6(t-a)-1$, and their total weight is $O(\sqrt{t}\log t)$. Consequently

$$R_{\mathrm{mid}}(t)\log(6t) \geq W_t - O(\sqrt{t}\log t) > 0$$

for all sufficiently large t by (36). □

Conversely, if $R_{\mathrm{mid}}(t) = 0$ then

$$\mathcal{C}_t = -3K_t + o(K_t) + O(\sqrt{t}\log t). \tag{37}$$

Thus whenever $K_t/(\sqrt{t}\log t) \to \infty$, a failure requires almost complete negative additive covariance. This is a structural *necessary condition for failure*, not proof that failures cannot occur.

The corresponding exact Fourier representation, with $F_t(\alpha) = \sum_{b=1}^{t-1} f(b)e^{2\pi i\alpha b}$ and $G_t(\alpha) = \sum_{a=1}^{t-1} g(a)e^{2\pi i\alpha a}$, is

$$R(t) = \int_0^1 F_t(\alpha)G_t(\alpha)e^{-2\pi i\alpha t}\,d\alpha. \tag{38}$$

It shows why controlling the frequency of transitions alone does not control their *additive alignment*. The binary prime configurations here are outside the nondegenerate setting of the Green–Tao linear forms theorem [6].

Proposition 10.2 (Second-order transitions are insufficient)**.** *There are two periodic binary sequences with identical frequencies of every length-three word, each with $f(16) = 0$, but with different values of $\sum_{a=1}^{15} f(a)f(a+1)f(16-a)$.*

Proof. Take sequences of period 16 whose first period is respectively

$$u = \texttt{0001011100101110}, \qquad v = \texttt{0001001011011110}.$$

Direct cyclic enumeration shows that every one of the eight three-symbol binary words occurs exactly twice in either sequence. With indices starting at 1, both have $f(16) = 0$; the two convolutions at 16 are respectively 0 and 2. □

11 Exact finite computations and stronger four-prime conjecture

Our supplied `verify_convolution.cpp` uses an exact number-theoretic transform modulo 998244353. Since every convolution coefficient under consideration is at most 10^6, reduction modulo this prime loses no information. The resulting statement is strictly finite.

Proposition 11.1 (Computational certificate, bounded range)**.** *For every $2 \leq t \leq 10^6$, $R(t) > 0$. There are exactly 793 498 indices in this interval with $f(t) = 0$, and none has $R(t) = 0$. The sample values are:*

t	2	92	1000	10^4	10^5
$R(t)$	1	16	47	238	1218

Furthermore $R(10^6) = 5608$.

An independent integer-loop computation of the restricted correlation finds no extinction-index failures for $2 \le t \le 10^5$; among *all* t in that range the sole restricted failure is $t = 38$, where $f(t) = 1$. Some values are

t	92	1000	10^4	10^5	10^6
K_t	8	29	211	1315	9279
$R_{\mathrm{mid}}(t)$	4	10	65	279	1321

The last entry was checked directly at that single t.

The even stronger four-prime question considered in our investigation is:

Conjecture 11.2 (Aligned sexy-prime Goldbach pairs)**.** For every sufficiently large n there is $d \ge 1$ such that

$$n - d - 6, \quad n - d, \quad n + d, \quad n + d + 6 \quad \text{are all prime.}$$

Setting $a = n - d - 6$ and $b = n + d$ shows that this asks for two lower sexy primes a, b with $a + b = 2n - 6$ and $b - a \ge 8$. A separate exact modular convolution (source included) gives exactly 38 exceptions for $12 \le n \le 10^6$, the largest at $n = 5804$. Thus the proposed threshold $n \ge 5805$ is supported on this finite range only. The stronger conjecture implies $I_3(n) \ne \varnothing$, but it is not used in our rescue proofs.

12 Conclusion and open proof obligations

The established chain is

$$\begin{aligned} &\text{modular extinction} \longrightarrow \text{exact convolution} \\ &\longrightarrow \text{positive local singular series} \\ &\longrightarrow \text{many rough candidates.} \end{aligned}$$

The missing arrow is from local candidate supply to a guaranteed *prime* complement. Complete-period CRT arguments cannot provide that arrow, and fixed-order transition statistics do not determine additive alignments.

The most focused sufficient conditions are the one-sided mixing hypotheses of Theorem 10.1. A stronger analytic approach would estimate the minor-arc contribution in (38) uniformly in t; a combinatorial approach would need a new invariant preventing all eligible sexy-prime positions from anti-aligning with the complementary prime positions.

Caution. We prove neither a uniform positive lower bound for $R(t)$ nor the existence of infinitely many sexy-prime pairs. The results of [9, 10, 1] concern related but distinct questions and do not imply our conjecture.

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